TIRust benchmark

Batched one-loop reduction: TIRust, IREGI, COLLIER and MadLoop

2026-10-24 · AMD EPYC 9754 · CPUs 0 / 254 · identified kernel snapshots

Reading guide. Every displayed timing ratio has TIRust in the numerator: *TIRust / IREGI*, *TIRust / COLLIER*, or *TIRust(candidate) / TIRust(baseline)*. Physical graphs use *TIRust / MadLoop*. Green means TIRust is faster, white parity and red slower. The archival reference-only table shows absolute times and outcomes. Gray is unmeasured; amber marks pending or inconclusive evidence. Source inspection does not establish numerical agreement. Measured TIRust6b24 passes eight numerical cases but retains seven timing regressions/one noisy row. It predates later APIs/preparation. Original-IREGI/COLLIER and physical sessions remain separate; private improvements, including the new unadopted lazy/eager cohort, do not close performance acceptance.

Acceptance gates

Correctness	Agreement with original IREGI and isolated MadLoop graphs	**Pending**
Kernel performance	Equal or better on identical inputs; separate measured snapshots	**Unmet**
Complete evaluation	Six physical rows measured; two failures retained	**Inconclusive**

1 Historical kernel baseline: `4c32ece`

Table 1: **Massive bubble/triangle, binary64, hybrid mode.** TIRust and IREGI total medians in µs/point. **TIRust / IREGI** ratios are paired-block medians; intervals are 99% bootstrap confidence intervals. Batch 1 is latency; batch 32 is amortized cost per point. Each backend uses its own paired reference.

Case	Rank	Interpreter Batch 1	Interpreter Batch 32	Native Batch 1	Native Batch 32
Bubble	0	3.176 µs IREGI 1.988 µs · ×1.598 99% CI 1.585–1.607	1.708 µs IREGI 1.885 µs · ×0.906 99% CI 0.902–0.911	0.936 µs IREGI 0.943 µs · ×0.993 99% CI 0.987–1.003	0.903 µs IREGI 0.878 µs · ×1.031 99% CI 1.024–1.042
Bubble	1	9.552 µs IREGI 5.707 µs · ×1.670 99% CI 1.655–1.688	5.831 µs IREGI 5.748 µs · ×1.015 99% CI 1.008–1.019	4.663 µs IREGI 3.044 µs · ×1.534 99% CI 1.526–1.549	3.231 µs IREGI 3.029 µs · ×1.069 99% CI 1.062–1.081
Triangle	0	29.152 µs IREGI 8.496 µs · ×3.446 99% CI 3.415–3.454	18.229 µs IREGI 8.433 µs · ×2.160 99% CI 2.147–2.172	4.227 µs IREGI 4.968 µs · ×0.849 99% CI 0.844–0.858	8.405 µs IREGI 4.899 µs · ×1.715 99% CI 1.704–1.723
Triangle	1	38.505 µs IREGI 18.297 µs · ×2.101 99% CI 2.088–2.108	25.634 µs IREGI 18.224 µs · ×1.408 99% CI 1.400–1.413	8.809 µs IREGI 10.418 µs · ×0.850 99% CI 0.844–0.854	12.461 µs IREGI 10.213 µs · ×1.220 99% CI 1.206–1.228



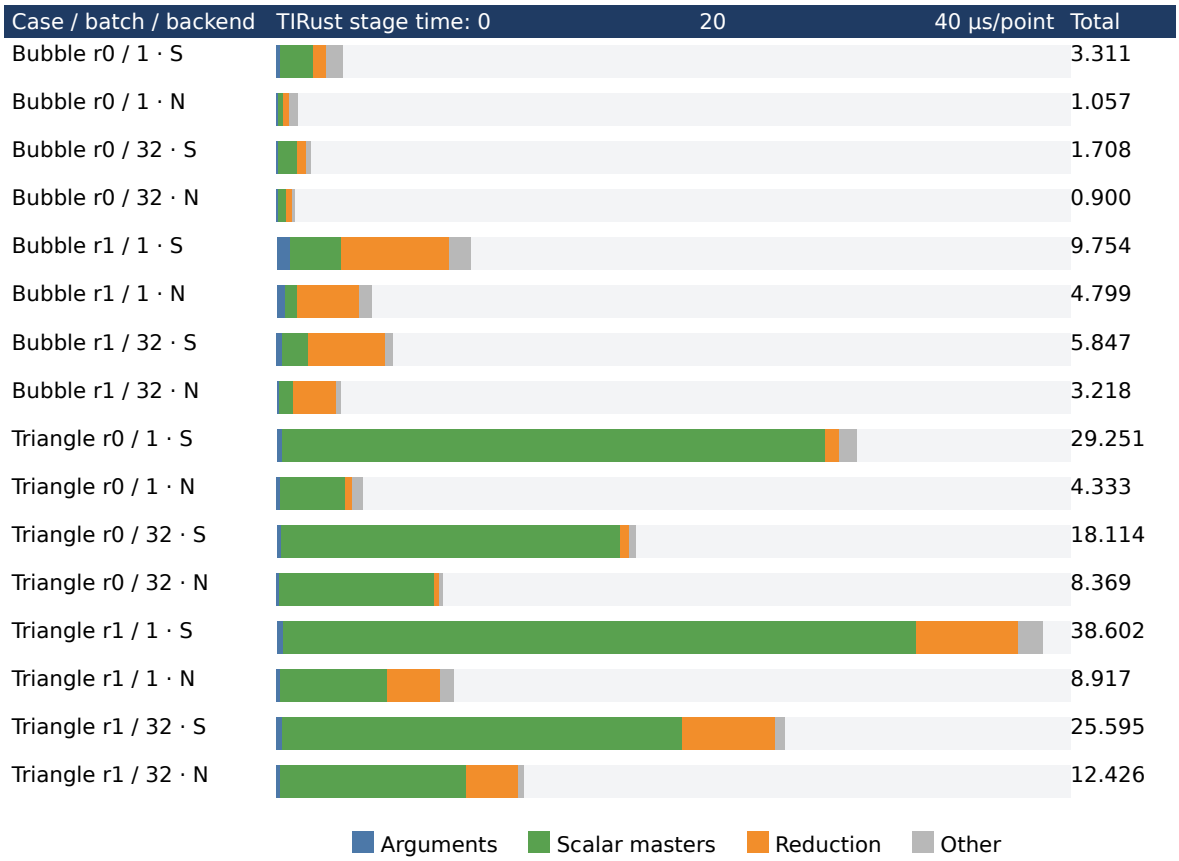
Both backends pass all eight numerical checks at 10^{-10}. The interpreter has seven regressions and one improvement; native has five regressions, one parity result and two improvements. **Absolute IREGI timings changed substantially between sessions**, despite identical inputs/reference binary and CPU settings. Each verdict uses its own paired run; no direct backend speedup is inferred across sessions. Both use the historical `4c32ece` revision; higher-rank performance remains unqualified.

The newer optimized interpreter snapshot `5c9e3dd` passes eight numerical checks, with six timing regressions and two noisy rows. The `80949db` snapshot gives five regressions/three noisy rows. Latest `6b24c03` passes all eight checks: seven regressions/one noisy row. Separate paired comparisons follow; performance acceptance is unmet.

Newer validation includes four integrated graphs. The top pentagon at Float 128/256 agrees with PV/hybrid and MadLoop; original IBP still fails. The latest physical timing comparisons remain inconclusive.

2 Historical stage profiles: `4c32ece`

TIRust medians from separate instrumented passes, in µs/point. All bars use 0–40 µs/point; S denotes the Symbolica interpreter, N the generated native backend. Sessions are separate: compare each with its own reference on page 1, not with the other backend's absolute bar length. Raw ns/batch are divided by 1000 × batch width.



"Other" is total median minus named stage medians, not a separately timed region. Scalar-master time includes complete master calls; reduction includes stability checks. IREGI has no measured stage split here. For native triangle rank 0, scalar masters cost 3.282 µs at batch 1, but 249.977 µs per batch 32, or 7.812 µs/point: this batch path has higher amortized master cost. The observation identifies a profiling target, not a cross-session speedup or a proven cause. Full stage values and samples remain in the raw data.

2.1 Preparation and measurement conditions

Backend	Bubble r0	Bubble r1	Triangle r0	Triangle r1
Interpreter	1.572 s	0.143 s	0.104 s	0.091 s
Native	0.011 s	0.012 s	0.001 s	0.013 s

Preparation values are single observations without repeated uncertainty, excluded from warm kernel ratios. Batch32 reuses the plan; numerator preparation is outside this tensor-array benchmark.

Both backends of the isolated pushed revision `4c32ece` ran on AMD EPYC 9754 CPU 0, with 5 warm-ups and 21 paired blocks targeting 30 ms, randomized order with seed 271828. TIRust used release opt-level3, thin LTO and one codegen unit; Fortran used `-O3`. Structural plans and allocation capacity were reused; numerical results were recomputed, and original recurrence caches were freed per sample. Subprocess transport is excluded. All source/binary provenance checks pass, and every ratio interval is no wider than 2% of its median. Raw samples, inputs, flags and hashes are retained in `validation/benchmark/baseline-4c32ece/` and `validation/benchmark/native-4c32ece/`.

A separate frozen-native policy experiment keeps `Whole`: all 18 C0/D0 comparisons across three orderings find `FixedRows4/8/16` slower, with per-comparison 99% intervals excluding parity. It uses 32 valid lanes, CPU 0 and 21 rotated blocks. This selects a batch policy, not IREGI speedup; see `validation/native-policy/`.

Native allocator adoption remains rejected. Historical 24 timing rows give two singleton parities/22 noisy rows; new D0 timings include two batch32 regressions. D0 exact old/new parity passes 40 configurations/1,488 complex outputs (f64/DD/Float256); interpreter differences remain. Separate candidate decisions appear in the optimization ledger and `validation/native-stack-batch/`.

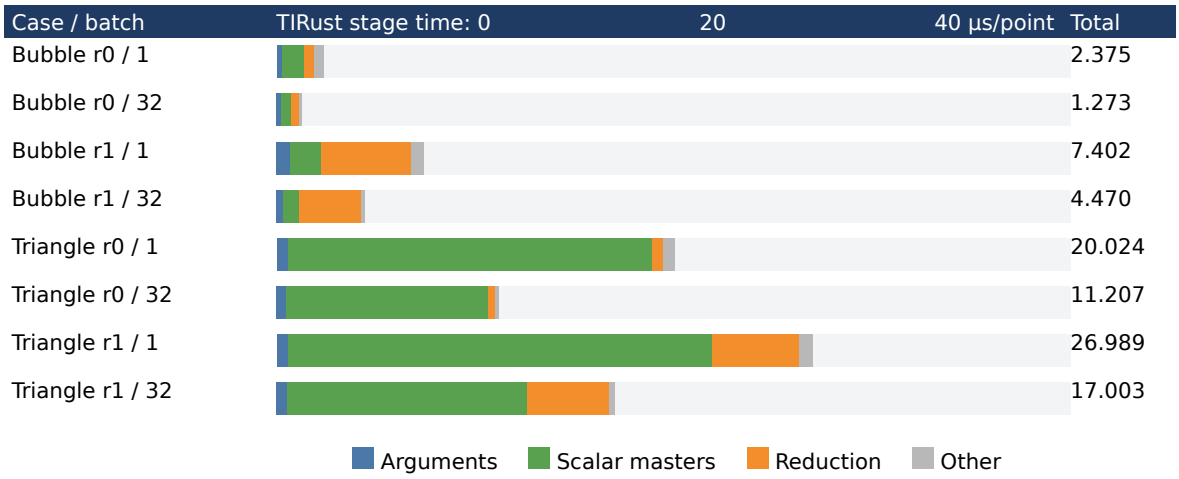
3 Latest kernel snapshot: `6b24c03`

All eight massive bubble/triangle checks pass 10^{-10}, with maximum normalized error 2.46×10^{-13}. Seven timing rows regress; the rank-one triangle singleton exceeds the unchanged 2% interval-width limit (2.357%) and remains **inconclusive**. This measured snapshot predates the standalone planar and finite-D0 certification APIs and later expression preparation. The performance requirement remains unmet.

Table 2: **Binary64 hybrid interpreter, µs/point.** Latency at batch 1 and amortized cost at batch 32. Ratios are **TIRust / IREGI**, using the same-run paired original-IREGI measurement.

Case	Rank	Batch 1	Batch 32
Bubble	0	2.225 µs IREGI 0.948 µs · ×2.344 99% CI 2.335–2.353 Regression	1.265 µs IREGI 0.896 µs · ×1.413 99% CI 1.407–1.416 Regression
Bubble	1	7.245 µs IREGI 3.048 µs · ×2.388 99% CI 2.372–2.403 Regression	4.498 µs IREGI 3.046 µs · ×1.480 99% CI 1.467–1.488 Regression
Triangle	0	19.618 µs IREGI 4.925 µs · ×3.986 99% CI 3.977–3.996 Regression	11.152 µs IREGI 4.963 µs · ×2.246 99% CI 2.237–2.255 Regression
Triangle	1	26.777 µs IREGI 10.356 µs · ×2.586 99% CI 2.570–2.631 Inconclusive: noise	16.967 µs IREGI 10.411 µs · ×1.627 99% CI 1.621–1.636 Regression

3.1 Separately instrumented component timings



Stages use a separate instrumented pass. “Other” is its total median minus named medians; stage sums are not the uninstrumented gate. Graph/numerator preparation is outside this tensor-array benchmark. The single plan-preparation observations remain in the data, not in speed ratios.

CPU 254, five warm-ups and 41 randomized paired blocks targeting 100 ms use identical inputs, complete tensor arrays and all native Laurent slots. Structural allocations are reused; numerical results are recomputed and reference caches cleared per sample. Transport is excluded. rustc 1.99 uses opt-level3, thin LTO and one codegen unit; GFortran 15.2 uses `-O3`.

The complete optimized workspace build exits successfully. All 414 immutable build-source hashes, both binaries, feature selection and harness match the retained provenance. Other project computations were held. CPU 254 was selected by a read-only load survey because unrelated work occupied CPU 0; external host activity remained uncontrolled. No system settings were changed. Historical sessions used different CPUs and timing protocols; only each session’s paired reference supports its verdict, with no causal revision speedup. Raw samples and reproduction: `validation/benchmark/baseline-6b24c03-licensed/`.

4 Direct TIRust / IREGI / COLLIER comparison

Private optimized prototypes compare the complete reduction call on the same massive Euclidean N2/N3 rank0/1 inputs. **B** is baseline TIRust with Symbolica masters; **S** is the same TIRust prototype selecting native A0/B0/C0 for plain f64. IREGI uses Hybrid and COLLIER uses COLI mode 1. Every ratio below has TIRust in the numerator; all forty results remain visible.

Table 3: **Same-cohort paired latency ratios and 99% CIs.** B and S denote TIRust variants, not separate reference libraries. Below one favors the numerator. Amber preserves the short-block family.

Family / batch	TIRust B / IREGI	TIRust S / IREGI	TIRust B / COLLIER	TIRust S / COLLIER	TIRust S / TIRust B
Bubble r0 batch 1	2.381 2.374–2.386 Short block	1.550 1.546–1.554 Short block	2.949 2.937–2.962 Short block	1.921 1.914–1.929 Short block	0.651 0.650–0.652 Short block
Bubble r0 batch 32	1.539 1.535–1.542 Slower	1.383 1.379–1.385 Slower	1.777 1.769–1.782 Slower	1.594 1.587–1.602 Slower	0.899 0.897–0.900 Gain
Bubble r1 batch 1	2.347 2.332–2.355 Slower	2.023 2.013–2.034 Slower	6.100 6.078–6.132 Slower	5.274 5.258–5.289 Slower	0.864 0.861–0.869 Gain
Bubble r1 batch 32	1.496 1.487–1.503 Slower	1.459 1.452–1.466 Slower	3.920 3.911–3.927 Slower	3.818 3.812–3.823 Slower	0.974 0.972–0.977 Gain
Triangle r0 batch 1	4.071 4.062–4.076 Slower	0.985 0.983–0.987 Gain	5.311 5.301–5.316 Slower	1.286 1.284–1.288 Slower	0.242 0.242–0.243 Gain
Triangle r0 batch 32	2.402 2.394–2.407 Slower	1.870 1.867–1.875 Slower	3.103 3.096–3.109 Slower	2.417 2.413–2.423 Slower	0.779 0.778–0.781 Gain
Triangle r1 batch 1	2.651 2.644–2.655 Slower	1.057 1.054–1.059 Slower	4.484 4.480–4.494 Slower	1.790 1.787–1.791 Slower	0.399 0.398–0.399 Gain
Triangle r1 batch 32	1.708 1.701–1.718 Slower	1.414 1.406–1.423 Slower	2.885 2.877–2.898 Slower	2.388 2.384–2.400 Slower	0.828 0.825–0.832 Gain

Selected versus COLLIER: seven regressions, one inconclusive family. Selected versus IREGI has six regressions, one gain (triangle rank 0 singleton) and one inconclusive family. Selected versus baseline has seven gains and one inconclusive family. Baseline/reference comparisons each retain seven regressions and one inconclusive family. No production/default adoption or complete performance acceptance follows.

Explicit posthoc analysis revision2. The original frozen analyzer marked five ratios short-block-inconclusive and 35 cohort-failed because one local short block set the aggregate status. A separately reviewed saved-data correction verifies clean termination and every original global check before assigning family gates. All forty sample arrays, medians, CIs, widths and numerical order-stratum statistics are unchanged; the conservative verdict remains archived. No new timing, threshold change, omitted family or repeated cohort.

One COLLIER block in bubble rank 0/batch 1 lasts 99.893971 ms, below 100 ms; all five ratios of that family remain inconclusive. Protocol: CPU 254, ten warm calls, 48 quartets per family containing all 24 permutations of the four engines twice, shared fixed iterations, 100 ms blocks, 10,000 paired bootstrap resamples, 99% CIs and 2% width gate. Every complete tensor/Laurent array and native pole probe passes before and after timing. Transport is excluded. This is a source-bound historical prototype, not the current production build. Sources: `validation/selected-native-four-engine/v3-frozen/` and `validation/selected-native-four-engine/posthoc-revision2/`.

5 Archival reference timings: IREGI and COLLIER

Identical massive Euclidean tensor integrals compare original IREGI Hybrid (mode 2) with COLLIER 1.2.10 COLI (mode 1). Each row requests the complete Cartesian tensor array and all four native Laurent slots. This is a reference-library comparison, separate from the TIRust performance gate. No TIRust/COLLIER ratio is inferred from different sessions. The direct four-engine cohort has its own separate source-bound table.

Table 4: **Archival binary64 reference timings, µs/point.** Batch 1 is latency; batch 32 is amortized cost per point. Qualified comparison outcomes are retained without reference/reference ratio displays.

Topology / rank	Batch	IREGI Hybrid	COLLIER COLI	Qualified outcome
Bubble rank 0	1	0.943	0.764	Inconclusive: short block
Bubble rank 0	32	0.927	0.766	COLLIER faster
Bubble rank 1	1	3.129	1.173	COLLIER faster
Bubble rank 1	32	3.122	1.172	COLLIER faster
Triangle rank 0	1	4.994	3.781	COLLIER faster
Triangle rank 0	32	4.963	3.825	COLLIER faster
Triangle rank 1	1	10.752	6.148	COLLIER faster
Triangle rank 1	32	10.864	6.145	COLLIER faster

The 99% interval width must be at most 2% of the paired median, and every block must reach 100 ms. Amber rows remain inconclusive. Raw ns/batch values are divided by 1000 × batch width for this display. Original paired ratios and intervals remain archival metadata in the JSON.

Numerical scope. All eight families pass the unchanged dimensionally normalized 1e-10 criterion; maximum error is 2.49e-13. The bubble rank0 singleton includes a 98.438 ms block below the 100 ms requirement and remains inconclusive; no row was rerun or omitted.

Cache policy. within-sample recurrence memoization in both; no cross-sample/event numeric reuse

COLLIER timing includes invariant setup and native conversion; IREGI includes mass setup and cache disposal. Both include the full tensor call and status checks.

Measurement protocol. CPU 254, five warm-ups and 41 balanced randomized paired blocks; common fixed iteration counts, registered minimum 100 ms. Both builds use GNU Fortran 15.2.0 -O3 without fast-math. Project computations were held; unrelated host activity remained uncontrolled. Transport and regulator probes are outside timing.

Ordinary non-dotted, fixed-dimension cases have zero IR poles; regulator probes are untimed. This archival timing cohort does not qualify higher points/ranks, singular or IR-divergent families or physical numerators. Raw observations and reproduction: `validation/collier-reference/timing-initial8/summary.json`.

6 Extended reference coverage: N2–N7

A separate correctness-only capture extends the ordinary, massive Euclidean reference matrix. Each cell keeps the native COLLIER admission status; numerical output from a rejected call cannot qualify a comparison. This table contains no additional timings and no new TIRust performance claim.

Table 5: **All 34 reference rows, including failures.** COLI-admitted means native status accepted and the complete native tensor/Laurent array agrees with IREGI within 10^{-10}. This table does not establish independent higher-point identities.

Propagators / rank	Batch 1	Batch 32
N2 / rank 0	COLI admitted	COLI admitted
N2 / rank 1	COLI admitted	COLI admitted
N3 / rank 0	COLI admitted	COLI admitted
N3 / rank 1	COLI admitted	COLI admitted
N4 / rank 0	COLI admitted	COLI admitted
N4 / rank 1	Unqualified	Unqualified
N4 / rank 2	Unqualified	Unqualified
N4 / rank 3	Unqualified	Unqualified
N5 / rank 0	COLI admitted	COLI admitted
N5 / rank 1	COLI admitted	COLI admitted
N5 / rank 2	COLI admitted	COLI admitted
N6 / rank 0	COLI admitted	Unqualified
N6 / rank 1	Unqualified	Unqualified
N6 / rank 2	Unqualified	Unqualified
N7 / rank 0	Unqualified	Unqualified
N7 / rank 1	Unqualified	Unqualified
N7 / rank 2	Unqualified	Unqualified

17 COLI-admitted, 17 unqualified rows. The six N5 rank 0–2 rows have maximum normalized disagreement 4.76×10^{-12}; the N6 rank 0 singleton reaches 2.60×10^{-13}. Existing N4 rank 1–3 rejections remain visible. Other higher-point calls retain COLI accuracy flags −1/−2 and DD error flag −10. Both policies and all raw statuses remain in the evidence; library code and rejection criteria are unchanged.

Attempt004 retained 239 validated isolated responses. Attempt005 appends only its eight missing requests, with no repeated capture or timing. The final N7 rank 2/batch 32 case retains unstable original PV, a nominally successful IBP result disagreeing with Hybrid by 0.520 normalized, and rejected COLLIER outputs. The N6 rank 2 mass-deformed IREGI disagreement of 1.97 also remains unresolved.

The independent saved-artifact/lane-gather audit verifies 81,086 assertions, 54,224 admitted native complex pairs and 5,427 rejected null slots. This is distinct from independent mathematical identities. IR-divergent, dotted and shifted-dimension COLLIER coverage is unqualified. Sources: `validation/collier-reference/shared/attempt004/` and `validation/collier-reference/shared/attempt005/`.

7 Independent checks and retained failures

7.1 Massive N5 reference identities

Three retained positive-F massive, undotted N5 branches at ranks 0–2 pass independent prefix, pinch/denominator/trace and cyclic/reversed-routing checks. Batch 32 cycles those same three branches; it does not add 32 distinct kinematic configurations. These synthetic checks use the reference codes, separately from TIRust reduction, physical numerators and MadLoop.

Independent check	Residual records	Complex slots	Maximum normalized error
Rank prefixes	924	3696	0
Pinch / denominator / trace	756	3024	3.135×10^{-12}
Cyclic / reversed routing	360	1440	4.101×10^{-12}
Finiteness / pole zeros	3357	10071	0

All 5397 residual records / 18,231 complex comparisons pass the unchanged 10^{-10} gate. The saved-data audit independently verifies 511,509 assertions and 346 archive members, including 465 admitted native lanes and 13,428 unique native complex pairs. The earlier capture's audit-pending status is immutable; the separate completed audit supplement supplies this result. No higher-rank, singular, dotted, dimension-shift, physical or timing qualification follows. Source: `validation/collier-reference/shared/identities006/correctness001/audit001/`.

7.2 Private N7: exact tests only

A frozen private N7 source build passes 17 exact tests. Its ignored generic variant-constructor test remains unrun. Source/build/test receipt review is distinct from numerical integral evaluation: no N7 full-array agreement, precision convergence, public adoption or performance is established. Source: `validation/highpoint-thin-n7-exact/`.

7.3 Private N6 rank 7: failed Float128 comparison

The retained Float128/requested-53 singleton returns 330 moments / 1320 native complex Laurent slots, including 990 exact zero pole slots. Its unchanged original-IBP comparison fails 90 finite slots: 12 at rank 6 and 78 at rank 7. Maximum normalized error is 2.539×10^{-9} against 10^{-10}. A completed process and reported stable status do not qualify the complete array.

For this real positive-F point, the independent analytic real-zero identity has no TIRust violations; original IBP and Hybrid each violate it in 88 slots. This explains 88 of the failed original-IBP complex comparisons from the reference real residual alone. It does not certify the remaining imaginary values or overturn the failed complete-array gate. Original PV stays explicitly unstable. Two saved quadrature witnesses are limited diagnostics, without rigorous error bounds or a causal cache-bug conclusion.

A separate Float256 diagnostic saves the same 1320 slots and 990 exact zero poles. Across Float128/256, the maximum observed normalized difference is 5.421×10^{-32}, below 2^{-80}. Original IBP, PV and Hybrid still disagree in 90, 151 and 109 slots respectively; unstable PV remains unqualified. This observed refinement is diagnostic: complete-array accuracy and precision-convergence acceptance remain false, with no high-rank adoption, lifecycle or timing result. The immutable saved-data packet and its initial missing-file failure are preserved at `validation/checkpoint-audits/2026-10-24-thin-n6-rank7-128/`. The separate terminal capture and independent audit are at `validation/highpoint-thin-n6-rank7-phase/`.

One-component quadrature diagnostic. For moment 126, q_0^6, six fixed-rule calculations (orders 12/16/20, two maps) give $(0, -0.05166894942216114)$. The estimated normalized uncertainty is 3.113×10^{-15} and is nonrigorous. Discrepancy plus that estimated margin is 3.249×10^{-15} for TIRust Float128, within the 10^{-10} diagnostic gate; original IBP and Hybrid give 4.274×10^{-10} and 4.267×10^{-10}, outside it. PV remains natively rejected. This supports one finite component; complete-array acceptance stays failed. Source: `validation/temporal-sixpoint-oracle/q0-sixth-001/qualification.json`.

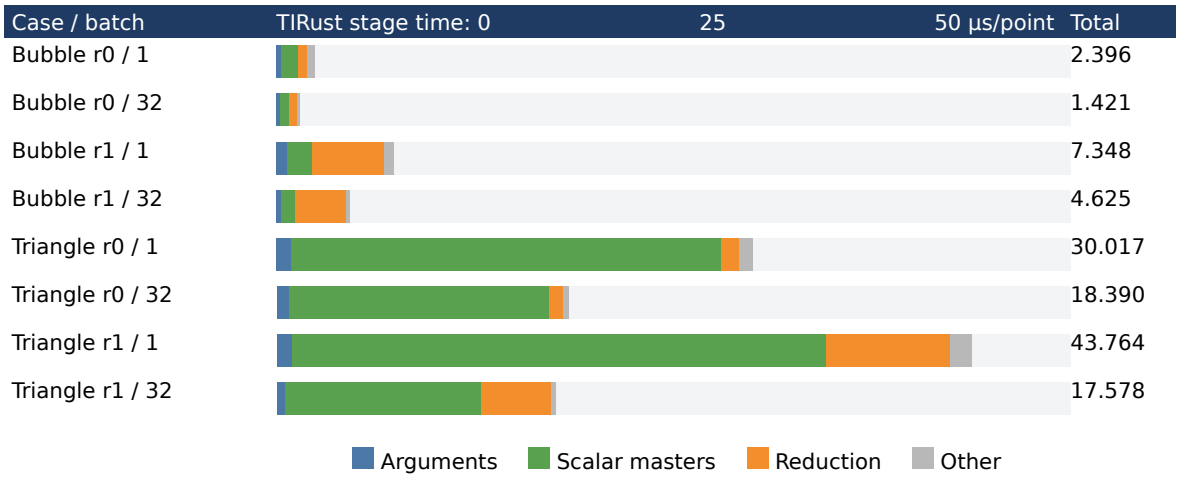
8 Historical kernel snapshot: `80949db`

All eight massive bubble/triangle checks pass the unchanged 10^{-10} criterion, maximum normalized error 2.46×10^{-13}. Five timing rows regress; three exceed the unchanged 2% interval-width limit and remain **inconclusive**. This optimized snapshot includes the adopted master stack allocation, generalized PV and lazy Cayley solve. Later core-allocation and experimental changes are excluded. The performance requirement remains unmet.

Table 6: **Binary64 hybrid interpreter, µs/point.** Latency at batch 1 and amortized cost at batch 32. Ratios are **TIRust / IREGI**, using the same-run paired original-IREGI measurement.

Case	Rank	Batch 1	Batch 32
Bubble	0	2.278 µs IREGI 0.946 µs · ×2.403 99% CI 2.389–2.420 Regression	1.380 µs IREGI 0.892 µs · ×1.550 99% CI 1.537–1.557 Regression
Bubble	1	7.243 µs IREGI 3.083 µs · ×2.342 99% CI 2.318–2.361 Regression	4.619 µs IREGI 3.077 µs · ×1.500 99% CI 1.490–1.518 Regression
Triangle	0	19.553 µs IREGI 4.967 µs · ×3.927 99% CI 3.911–3.944 Regression	17.604 µs IREGI 7.296 µs · ×2.374 99% CI 2.325–2.432 Inconclusive: noise
Triangle	1	43.420 µs IREGI 17.458 µs · ×2.500 99% CI 2.463–2.578 Inconclusive: noise	28.540 µs IREGI 17.009 µs · ×1.672 99% CI 1.490–1.762 Inconclusive: noise

8.1 Separately instrumented component timings



Stages use a separate instrumented pass. “Other” is its total median minus named medians; stage sums are not the uninstrumented gate. Graph/numerator preparation is outside this tensor-array benchmark. The single plan-preparation observations remain in the data, not in speed ratios.

CPU 0, five warm-ups and 21 randomized paired blocks targeting 30 ms use identical inputs, complete tensor arrays and all native Laurent slots. Structural allocations are reused; numerical results are recomputed and reference caches cleared per sample. Transport is excluded. rustc 1.99 uses opt-level3, thin LTO and one codegen unit; GFortran 15.2 uses `-O3`.

All 401 immutable build-source hashes, both binaries and the harness match the retained provenance. Other project computations were held; a pending native compiler was suspended. Host activity outside the project was not controlled. The accepted license contains no tracked key material. This is a separate session from the historical tables; its absolute timings establish no revision speedup. Raw samples and reproduction commands: `validation/benchmark/baseline-80949db-licensed/`.

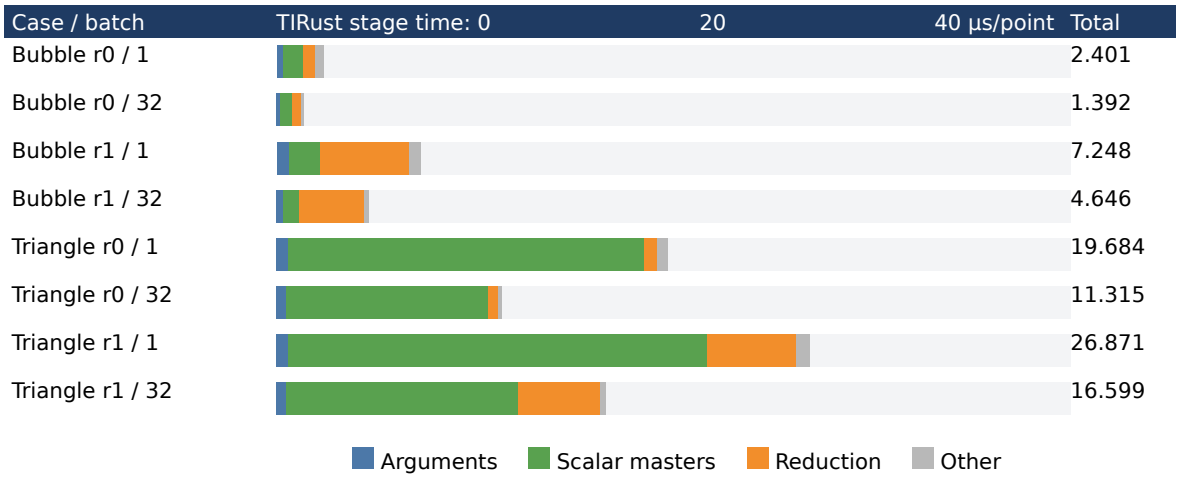
9 Historical kernel snapshot: `5c9e3dd`

All eight massive bubble/triangle checks pass the unchanged 10^{-10} criterion. Six timing rows regress; two exceed the allowed 2% relative confidence-interval width and remain **inconclusive**. Amber cells denote noise, not improvement. This commit predates default stack reuse, generalized PV and the vector-solve candidate. The overall performance requirement is unmet.

Table 7: **Binary64 hybrid interpreter, µs/point.** Latency at batch 1 and amortized cost at batch 32. Ratios are **TIRust / IREGI**, using the same-run paired original-IREGI measurement.

Case	Rank	Batch 1	Batch 32
Bubble	0	2.271 µs IREGI 0.955 µs · ×2.381 99% CI 2.364–2.393 Regression	1.389 µs IREGI 0.901 µs · ×1.541 99% CI 1.535–1.546 Regression
Bubble	1	7.146 µs IREGI 3.019 µs · ×2.367 99% CI 2.362–2.384 Regression	4.593 µs IREGI 3.020 µs · ×1.521 99% CI 1.516–1.532 Regression
Triangle	0	19.619 µs IREGI 4.975 µs · ×3.948 99% CI 3.932–4.027 Inconclusive: noise	11.310 µs IREGI 4.948 µs · ×2.281 99% CI 2.263–2.296 Regression
Triangle	1	26.724 µs IREGI 10.231 µs · ×2.610 99% CI 2.597–2.624 Regression	16.733 µs IREGI 10.279 µs · ×1.620 99% CI 1.608–1.649 Inconclusive: noise

9.1 Separately instrumented component timings



The stage sum belongs to a separate instrumented pass; "Other" is its total median minus named medians. These bars locate costs, not differences between revisions. Preparation is a single observation: bubble r0/r1 0.822/0.112 s; triangle r0/r1 0.067/0.061 s. It is excluded from the warm kernel ratios.

CPU 0, five warm-ups and 21 randomized paired blocks targeting 30 ms use identical inputs and all requested tensor slots. Structural allocations are reused; numerical results are recomputed. Transport is excluded. rustc 1.99 uses opt-level3, thin LTO and one codegen unit; GFortran 15.2 uses `-O3`. All 391 source hashes, both binaries and the harness remain unchanged under the retained provenance checks. Absolute timings from earlier sessions cannot establish a revision speedup. Raw blocks, inputs, manifests and reproduction commands: `validation/benchmark/baseline-5c9e3dd/`.

10 Physical pipeline measurements: `80949db`

Six preserved-input binary64 comparisons pass complete canonical tensors against all three original modes and all three isolated graph Laurent slots against MadLoop. **Every measured timing row remains inconclusive**: its 99% interval exceeds the unchanged 2% relative-width threshold. Both top-pentagon rows fail during preparation; no output or timing substitutes for them.

Table 8: **Equivalent complete numerical pipelines, ms/point, TIRust / MadLoop.** Repeated captured point; amortized batch32 cost does not establish throughput over varying phase-space samples.

Graph	Ranks TIRust/ML	Batch 1	Batch 32
Top triangle	2 / 3	0.877 ms MadLoop 0.108 ms · ×8.114 99% CI 7.808–8.575 Inconclusive: noise	0.729 ms MadLoop 0.105 ms · ×6.918 99% CI 6.817–7.061 Inconclusive: noise
Top box	4 / 4	11.689 ms MadLoop 0.971 ms · ×12.036 99% CI 11.854–12.303 Inconclusive: noise	5.088 ms MadLoop 0.978 ms · ×5.203 99% CI 5.087–5.421 Inconclusive: noise
Massless up box	4 / 4	7.242 ms MadLoop 0.904 ms · ×8.008 99% CI 7.642–8.225 Inconclusive: noise	3.620 ms MadLoop 0.888 ms · ×4.075 99% CI 3.997–4.116 Inconclusive: noise
Top pentagon	4 / 5	Failed before timing Preserved-input conditioning guard	Failed before timing Preserved-input conditioning guard

10.1 Scope and uncertainty

The timers include numerical numerator coefficients, reduction, conversion, contraction, status checks and returned-buffer/cache cleanup. Graph/model and external-wavefunction preparation, input packing, final comparison adapters and process transport are excluded from both. Bare poles and R1 are retained; R2 and UV counterterm graphs are excluded. Routing and rank differences are declared algorithmic choices: **no tensor-stage timing ratio is valid**.

Both engines use Hybrid, 53 working/requested bits, input preservation, CPU 0, five warm-ups and 21 blocks targeting 30 ms. Separate engine runs use an independent bootstrap ratio of medians (10,000 resamples). These intervals describe within-run variation and do not bound drift between engines. Project computations and producer writes were frozen; external host activity was not controlled. Stage samples and raw full responses remain available.

Attempt002 preserves all source/build identities. Attempt001 is retained as unqualified because producer inputs changed during execution. The optimized top-box numerical pass is specific to this captured harness path; earlier CLI binary64 guard rejections remain part of the ledger. Top-pentagon failure is the actual preserved-input derived-distance conditioning diagnostic. No failure is removed from the family count.

Sources, contracts, stage timings and reproduction: `validation/benchmark/physical-joint-quiet/`. The preceding three-block ggH smoke and 80 Python boundary/provenance tests are separately retained in `validation/benchmark/physical-joint-smoke/`.

10.2 Precision follow-up on the failed pentagon point

Promoting the same 95 represented binary64 input components exactly to Float256/512, with requested accuracy53 unchanged, gives convergence over all 280 complex tensor slots within 2.17×10^{-55}. Original PV/hybrid agree within 4.62×10^{-14} and the preserved MadLoop graph within 9.29×10^{-14}. Original IBP still differs by 0.60013; binary64/128 guard failures remain. The first 256-bit driver captured valid outputs before a rational-input comparison-parser failure; the retained separate comparison passes. This is one-point manual precision qualification, not automatic rescue or a timing gate. See `validation/physical/gghhh-guard/`.

11 Current implementation checkpoint

Production snapshot `06706e2` and private experiments have separate source and binary identities. Correctness evidence does not replace timing gates.

11.1 Ordinary scalar-master storage

The default interpreter now reuses root temporaries through Symbolica's existing allocator. All five families preserve every output, precision and signed zero in 40 configurations each across f64, DoubleFloat and Float256. Shared helpers and Dual preparation are unchanged.

Root temporary slots	A0	B0	dB0	C0	D0
Before	38	503	1,951	220	25,010
After	6	13	24	8	25

The default-stack workspace snapshot compiled with unchanged source hashes before the generalized-PV promotion below. Its current-guard acceptance passes 30 execution, 29 stability, six public mixed-precision and three scalar integration tests.

The optimized `80949db` snapshot also passes eight complete bubble/ triangle kernel checks at batches 1/32, maximum normalized error 2.46×10^{-13}. Three short blocks, zero warm-up and concurrent builds make this a correctness smoke only; its recorded timing ratios are unqualified.

The paired DD E6 experiment preserves all three statuses and 12 complex Laurent outputs exactly, with the independent 10^{-23} check passing in both variants. Each uses order 128 and 3,720 D0 rows.

Diagnostic observation	Before	After
Whole validation duration	46.682 s	43.822 s
Per-child peak resident memory	2,052,724 KiB	558,752 KiB

These are single-run diagnostics, with other compilation active; no timing gate or speed ratio is inferred. The paired experiment uses an earlier, stricter Float guard; fixed-precision DD is unaffected. Current-guard tests are separate. Sources: `validation/stack-reuse/production-factory/`.

11.2 Generalized PV and unresolved precision

The retained original-IREGI dotted-integral comparisons cover positive propagator powers and dimension shifts; they are separate from COLLIER's ordinary undotted slice. Generalized PV for $N \leq 4$ passes nine focused tests and 36 original-IREGI comparisons across 312 complex slots; maximum normalized error is 7.04×10^{-12} at the unchanged 10^{-10} gate. The promoted PV and lazy-solve core also passes five supplemental behaviors: supplied products, multinomial weights, complex scalar/tensor comparisons and Hybrid isolation. The latter uses working128/256, requested53; its binary64 fixture failure is retained. These checks use frozen former-default master archives; combined default-stack integration is separate. Higher points retain IBP/E6. Sources: `validation/generalized-pv/` qualification, promotion and supplemental manifests.

The exact small-routing E6 oracle exposes a remaining production failure:

Working bits	Requested bits	Public result
128	128	Unstable: nonfinite box output
256	128	Unstable: nonconvergent quadrature

Both runs finish within their bounds and expose no successful values. No oracle comparison or requested-128-bit agreement is claimed. The frozen binary predates default stack reuse. See `validation/shifted-pentagon/production-series-qualification.json`.

12 Validation coverage

Table 9: **Evidence ledger.** “Source audited” records code inspection; “verified” identifies executed checks described in that row.

Capability	Required evidence	Status
Independent reference oracle	Untouched/repaired Fortran twins 24 ordinary cases identical; 80 table entries exact; 7 generalized fixtures pass. New pentagon audits reveal reference defects.	Verified
Scalar-master component	OneLoopMaster A0/B0/C0/D0/DB0 Final27 scalar checks and13 consolidated D6 checks pass; all5 mixed physical D0 cases pass atFloat384. Unsupported derivative domains remain explicit.	Verified
IBP component	Powers, dimension shifts and singular identities 21 exact tests; the independent Gamma oracle covers 200 tensor requests and 40 generalized requests in all four slots.	Verified
PV component	Every tensor moment and native Laurent slot Canonical N1–4 ranks 0–7 pass. Generalized N≤4: nine focused + 36 original comparisons pass. Saved N5 rank-seven baseline: 1320 slots. Public N6/N7 tensor ranks pending.	Partial
Precision and Dual	binary64, DoubleFloat, Float, static/dynamic jets Selected f64/DD/Float128–512 and static/dynamic jets pass. Mixed-precision guards: 29 + 6 pass. Small-routing E6 requested at 128 bits: Unstable.	Partial
Execution / CLI / core oracle	Batches through reduction and scalar masters 30 execution and5 argument-guard tests pass. CLI19 smoke checks pass; frozen31 fixtures include30 binary64 plus1 explicit verified rescue.	Partial
Exceptional kinematics	Explicit errors and verified fallback N5 rank-six rescue passes; saved rank-seven baseline passes 1320 slots. Legacy: 22 primary + 8 rescues. E6 failures and broader domains remain visible.	Partial
Graph/numerator preparation	FeynKit DOT to Symbolica coefficients 4 model-boundary, 13 preparation, 4 physical numerator and 19 CLI smoke checks pass.	Verified
Physical MadLoop comparison	Graph-isolated top loops and massless box Integrated ggH, ggHH, ggHHH and massless up-box pass their retained MadLoop comparisons. ggHHH Float128/256 also agrees with original PV/hybrid; original IBP still disagrees and all-mode reports remain failed.	Partial

Snapshot boundary. Counts describe the executed component, public-execution and selected physical checks below. They do not certify all kinematics, higher-point ranks or every physical graph. Open discrepancies remain listed with the original tolerance. Four physical triangle and five massive/complex box cases pass six-component mixed derivative checks. Historical timeouts remain visible. The N5 rank-six public rescue verifies all 210 moments / 840 slots at 128/256 bits, including an independent coefficient. Binary64 TIRust and original IBP still exceed the fixed 10^{-10} criterion for that point. The frozen oracle passes 31 cases: 30 at binary64 and one generalized triangle through verified rescue. Broader domains remain pending.

12.1 Known original-reference defect

For shifts $(0,0,0,0)$ and $(0,1,0,0)$, squared masses $(4,9)$, $\mu = 3$ and numerator q_x^4, original PV and hybrid report stability but disagree with an independent Feynman-parameter integral. Their UV residues agree.

Method	Finite coefficient / i
Parameter integral	45.4316300015058951832997…
Original IBP	45.43163000150602
Original PV / hybrid	46.11123944517427

Evidence: `validation/pv-high-rank/evidence.json`. The original PV branch is retained unchanged; the relevant TIRust comparisons use original IBP and the independent parameter-integral result.

The dedicated original-IREGI/COLLIER table covers ordinary, non-dotted N2/N3 rank0/1 inputs, batches1/32. Its numerical and timing results do not qualify broader N4–N7, singular or IR-divergent families.

13 Measurement protocol

Pin release builds to one CPU core and record thread counts, hardware, OS, toolchain, flags, dependency/input hashes, seeds and warm-up. Identify cold or warm caches. Retain raw repeated-block observations, median, dispersion and an output checksum.

Preparation includes graph import, numerator construction and evaluator compilation. Time numerator, reduction, masters and total independently; summed instrumented stages are only diagnostic. Separate library costs from subprocess/file I/O and match physical routing, modes and stage boundaries.

Use identical samples at batch widths 1, 4, 8, 16, 32 and 64, matching numerical precision and including rescue costs. Retain failed lanes. Every speed gate requires correctness: stable binary64 uses the fixed 10^{-10} mixed tolerance on dimensionally normalized outputs. Gate singleton latency and batch-32 throughput per topology/rank/mass family, with IBP, PV and hybrid separate. The target envelope includes seven propagators, rank seven, generalized powers and dimension shifts.

The kernel harness now records every requested configuration before starting subprocesses, preserving interrupted and unrun rows. The latest Python suite passes 80 ledger, provenance, protocol and comparison checks. The optimized joint ggH smoke passes at batches 1/3; its short zero-warm-up timings are diagnostic. The subsequent quiet physical results are reported separately.

Private slim RealBall passes 88 core checks, eight kernel and 15 typed cases, two ggH packets, and 10 separate planar-interaction checks with retained dependencies. Its first four timing rows give two noisy real-lane results, one complex-lane parity and one gain. The longer cohort is reported separately; the candidate remains unadopted.

With the accepted license, 360 batched calls across two workers match sequential A0–D0 values, status and metadata for f64/Float128 and second-order Dual branches. A separate four-worker cold-specialization test passes shared ownership and cleanup checks. These qualify retained SymbolicaBatch snapshots; native kernels remain separate. The earlier rank-one box rejection is retained in `validation/concurrent-workspaces/`.

14 Rebuilding this report

Update `docs/data/results.json` only from retained measurements and rebuild with `bash scripts/build-benchmark-report.sh`. The report uses Libertinus Serif and DejaVu Sans/Mono. Supply their installation directories through Typst's `--font-path` option when needed. Reference checkout revisions are retained in the JSON data. They record the audited source trees and must not be mistaken for a verified dependency build.

14.1 Validation reproduction and remaining checks

Run recorded commands with the pinned environment from `scripts/env.sh`. Exact commands and source hashes accompany each JSON result. The frozen 31-case oracle includes one explicit 128/256-bit rescue; all 30 legacy fixtures pass through 22 primary results and eight verified rescues. Three active rescue, four bubble-Taylor, three scalar API and nineteen CLI smoke checks pass. The new 68-test default-stack acceptance is a separate source scope; earlier failures remain retained.

The final frozen master source passes 27 checks, including extreme-scale B0, six-component scale-only jets in all families, mixed input precision and the full physical D0 derivative batch; see `validation/scalar-masters/qualification.json`. The separate exact factory-parity/default-stack qualification is described above. Generated native kernels remain opt-in; no new timing gate passes.

An independent Symbolica Gamma expansion checks 200 tensor and 40 generalized requests with analytic bubbles disabled/enabled: 480 evaluations and 1920 slots. Nine extracted FeynKit partial-fraction and two planar formula tests pass; the separately qualified explicit planar API is recorded below.

N6 inline 128/256-bit arrays converge over 1320 slots within 6.46×10^{-29}; public N6/N7 tensor arrays remain unqualified. Adopted alias-use collection preserves 29 paired cases, 86 successful lanes and 4520 components. Frozen collector-plus-sum and private amplitude-variable probes reach the 600 s bound (601.232/601.329 s with teardown), after primal amplitudes but before master activations finish; neither produces arrays. The latest private variant passes 48 variable-set, 34 amplitude/error and 73 semantic checks. Earlier timeouts remain in `validation/public-highpoint/` no timing gate follows.

15 Physical fixture qualification

Single graph	Pointwise numerator	Integrated TIRust result
Top triangle: $gg \to H$	Verified	Verified
Top box: $gg \to HH$	Verified	Verified
Top pentagon: $gg \to HHH$	Verified	PV/hybrid + MadLoop
Massless up-quark box	Verified	Verified

15.1 Physical reference contract

Fixtures preserve parameters, actual HELAS polarizations, graph factors and source hashes. The triangle uses two orientations with grouping disabled. MadLoop shifts are $(0, p_1, p_1 + p_2)$ with polynomial variable $x = -Q$; its adapter includes the odd-rank sign. For the FeynKit routing, $N_{\mathrm{FK}(q)} = -\frac{1}{2} P_{\mathrm{ML},1}(p_1 - q)$ with matched couplings/polarizations. Do not apply color or direction factors twice.

Five captures contain 40 graph records, including a second triangle helicity; these are not 40 integrated validations. Thirteen preparation and four selected physical numerator comparisons pass. Actual evaluated momenta are exported because MadLoop may correct the requested point.

Fresh preserved-input Fortran replay matches all 301 numerator coefficients, 903 tensor Laurent entries and 12 contracted slots across four graphs exactly; eight alignment boundary tests pass. Legacy-default ggH is unchanged. This is reference-only qualification; later TIRust checks belong to the separate joint run. Equivalent full-graph timing retains TIRust/MadLoop ranks 2/3 for ggH, 4/4 for both boxes and 4/5 for ggHHH, and their declared routing. It does not qualify tensor-stage ratios. Sources: `validation/benchmark/physical-alignment/`.

The recorded binary64 $gg \to H$ CLI passes all 15 moments / 60 slots against original IREGI within 1.86×10^{-15} normalized error and MadLoop within 9.38×10^{-15}, including poles. Values and binary identity remain in `validation/physical/ggh-cli-comparison.json` this is not a fresh replay. Nineteen CLI smoke checks pass. Binary64 comparison arithmetic certifies no extra digits from higher working precision; short unpinned timings are excluded.

The historical top-box binary64 CLI rejects: PV fails its cancellation guard and IBP fallback is nonfinite. Float128/256 reruns pass all 280 slots against all three reference modes within 1.15×10^{-13}, and MadLoop within 4.77×10^{-15}. Each freshly prepares/evaluates the numerator. Their normalized tensor difference is 6.09×10^{-37} under binary64 comparison arithmetic. See `validation/physical/gghh-cli-precision-convergence.json`. The Symbolica-backed model adapter passes four boundary, thirteen preparation and four numerator tests while preserving other numerical tokens.

The 80949db massless up-quark box passes all 70 moments / 280 slots against each original mode: maximum error 2.85×10^{-14} against IBP and 7.11×10^{-15} against PV/hybrid. MadLoop finite/single/double coefficients agree within 6.74×10^{-15} with fixed-color factor $-\frac{1}{8}$. Its one-run wall time is unqualified; see `validation/physical/gggg-up-cli-comparison.json`.

Complete top-pentagon CLI reruns at Float 128/256 now agree with original PV/hybrid for all 280 slots within 4.63×10^{-14}, and with MadLoop within 9.30×10^{-14}. Between-run tensor difference is 7.79×10^{-17} under binary64 comparison arithmetic. Original IBP still differs by 0.60014, so the all-mode report remains failed. See `validation/physical/gghhh-cli-precision-convergence.json` and linked reports. Earlier PV/hybrid replay of 378 captured MadLoop slots was exact.

15.2 Physical derivative qualification

Four triangle cases at 384 bits cover real subthreshold/absorptive and complex-mass kinematics, each with six first, second and mixed Dual components. Finite differences converge with maximum final scaled error 5.47×10^{-28}. One equal-complex D0 case also passes, with error 1.11×10^{-34}. The complete five-case D0 batch now passes at 384 bits, maximum final scaled error 2.53×10^{-30} under the unchanged 2^{-70} criterion. Its 224.83 s validation duration is not a benchmark; earlier 180 s timeouts remain retained. See `validation/scalar-masters/physical_box_full.json`. Exact commands and scope are retained in `crates/iregi-masters/tests/validation_evidence.json`.

16 Open qualification ledger

Unresolved item	Observed evidence and remaining check
Binary64 baseline	Historical TIRust/IBP errors 2.23×10^{-10} / 3.40×10^{-10} fail. Rank-six rescue passes at the retained point.
Pentagon coverage	Rank-six rescue passes 840 slots; saved rank-seven baseline passes 1320. Later precision/routing edits and broader domains require qualification.
Physical pentagon	Float 128/256 agrees with PV/hybrid and MadLoop; original IBP disagrees. Binary64 and all-mode acceptance remain incomplete.
Shifted scalar	Thirteen standalone checks and public f64 fixtures pass. Production small-routing E6 remains Unstable at requested 128. New private mixed-jet checks cover common masses/zero invariants only; broader domains and production integration remain unqualified.
Derivative domains	Five-case mixed D0 checks pass 2^{-70}. Thresholds and unsupported transverse massless/Gram/Cayley domains retain errors.
Physical top box	Historical binary64 guard rejection remains visible; Float 128/256 and preserved 80949db binary64 checks pass at retained points.
Supplied-product N6	Certified scalar working 128/256, requested 53 passes with zero poles and invalid-lane isolation; binary64 rejects. Independent difference 1.53×10^{-23} is below estimated (nonrigorous) 6.11×10^{-22}. N7/tensors/mixed Dual remain pending.
Legacy and precision	All 30 retained cases pass: 22 primary and 8 rescues. Three false-success cases are repaired; conservative active-master error estimates remain noncertified. Broader domains remain unqualified.
Performance	Historical 4c32 interpreter: 7 regressions/1 gain; native: 5 regressions/1 parity/2 gains. 5c9: 6 regressions/2 noisy; 80949: 5/3. Latest 6b24: 7/1. Physical: six noisy/two failures. Overall acceptance is unmet.

16.1 Optimization and experimental evidence

Candidate	Executed evidence and decision
Adopted allocation / guards	Core allocation: 67 current opt2 checks; quiet paired run preserves 92 outputs. Guard context: 76 checks and 120 snapshots, 14 cleanup checks; six isolated gains/six noisy rows. Combined 6b24 optimized IREGI gate remains seven regressions/one noisy row.
Roundoff cache	29/37 stability checks and 120 snapshots pass. Candidate rejected: two regressions/ten noisy rows. Separate accuracy-guard diagnostic: two regressions/six noisy rows.
CLI coefficients	72 ordinary/12 mixed batches pass. Eight opt2 timing rows: five gains/one parity/two noisy; ggHH32 ratio 1.0367, CI 1.0237–1.0501 retains possible regression; unadopted. CSE proves 892 coefficient/223 homogeneity identities. 1281 raw failures coexist with zero dimension-normalized diagnostic failures; no physical acceptance or new timing claim.
Native candidates	Dense f64 B0/C0: two gains/six singleton parities/four batch32 regressions. D0 allocator: three singleton gains/two batch32 regressions/one noisy. Shared B0/C0 kernels: one gain/two parities/seven regressions/two noisy. All rejected; D0 singleton source is unchanged. Separate actual A0–C0 replay passes 1782 outputs from 297 rows at 10^{-10}; native/interpreter values are not bitwise identical. New replay timings are reported separately.
Certificate candidates	Slim RealBall module: 23 checks, including 8 baseline controls. Separate exact-distance receipts: 49 checks and 120 exact snapshots against the same private RealBall baseline; argument rows unchanged. Fourteen timing cases give eight gains/two parities/four noisy, no qualified regression. Private final integration passes 12 receipt/6 accounting checks, counted once with the automatic-planar followup. The final timing cohort is reported separately.
N5 planar API	Promoted explicit unit-power API: rank7 ordinary128/256 passes 1320 slots/convergence; mixed128 passes 5280 slots per accepted lane versus256 boxes, requested53. Transverse jets reject; 9 focused/49 ordinary checks and 20 Fortran-leaf coefficient cases pass. Earlier timeout remains. Private automatic rank1/2 dispatch passes 69 checks; the final 18-test followup is separate. No automatic rank7, production fallback or timing claim.
Finite D0 certification	Explicit API promoted: three actual public tests and current core Cargo checking pass. Exact represented inputs receive component-budget enclosures in the checked near-vacuum domain; upstream routing and quadrature errors are excluded. Prior typed targets are f64/40, DD/80, Float128/100 and Float256/128. Ordinary evaluation and default E6 remain unchanged; no automatic certified-master dispatch.

17 Latest isolated optimization cohorts

These private candidates were measured in one coordinated quiet window on CPU 254. Each has its own frozen sources, binaries and registered protocol; results do not qualify a combined implementation or replace the TIRust/IREGI performance gate. Candidate effects are never multiplied. Failed controls, inconclusive rows and earlier cohorts remain in the evidence.

Private cohort	Per-family outcomes	Controls, scope and decision
Final proof receipts	14 families: 11 gains, 1 parity, 2 regressions; no noisy intervals.	Ordinary bubble singleton regresses 2.90% (CI 2.43–3.27%); ordinary triangle batch32 regresses 0.69% (CI 0.20–1.26%). No blanket adoption. Same retained masters; N5 path inactive. Earlier cohort remains separate.
Slim RealBall, longer run	4 families: 1 gain, 1 noisy, 2 regressions.	Real singleton improves; real batch32 remains noisy (CI width 2.18%). Complex-mass controls regress 0.74%/0.62% at batches1/32. Real shifts remain; no adoption or pooling with the first run.
Native master replay	8 families: 8 gains; all 168 pairs faster.	TIRust(candidate)/TIRust(baseline) ratios 0.143–0.809 on actual A0–C0 master bundles: native candidate, Symbolica baseline. Raw kernels exclude adapter validation/allocation, recurrence and contraction. No production switch or IREGI parity claim.
CLI coefficient CSE0	8 families: 8 gains.	Complete warm coefficient call only. Raw coefficient criterion still fails in baseline and candidates; separate dimension-normalized diagnostic passes. No adoption or physical acceptance claim.
CLI coefficient pair1	8 families: 6 gains, 2 noisy.	ggHHH singleton and up-quark box batch32 remain inconclusive. Preparation observations are retained separately; repeated fixed-point f64 evaluation is not a full-graph timing gate.

Ratios in these cohorts are **TIRust(candidate) / TIRust(baseline)**, using each cohort's own implementation and baseline. Every measured family is retained in the linked JSON, with raw paired observations, 99% confidence intervals and unchanged qualification thresholds. Project compute was held, but unrelated host activity remained uncontrolled. Earlier measurements use different frozen sources or protocols and remain separate. No new production adoption is implied by this page.

17.1 Exact and source-only research

Additional Symbolica CSE leaves A0/B0/C0 exports unchanged: 43/506/223 instructions and 6/13/8 root slots. Eighteen root identities and 470 control checks pass under the structured-SSA assumption. With no instruction saving, the candidate stops before numerical or performance qualification.

The E6 design notes remain proposals. The initial private direct N2 prototype at Float256/request128 certifies 16 components across four successful lanes with one-axis common-mass jets; five expected rejections hide values. Independent audit: 513 checks and 140 hashes pass. Its later typed followup appears separately below. Neither snapshot qualifies public/default E6, general geometry or performance.

A separate private rank-lowering N6 rank-one candidate passes all 20 complex Laurent slots against original IBP at Float128/request53, maximum normalized error 3.17×10^{-13}. The old public path rejects nonfinite coefficients, so this is not paired old/new numerical agreement. Exact constraint tests accompany the frozen development build. Its rank-two probe rejects nonfinite coefficients in 57.823 s, within the 180 s bound, without arrays. Rank-seven, N7, convergence and performance qualification remain pending.

18 Lazy proof capture: complete paired comparison

TIRust(lazy) / TIRust(eager) measures the entire reduction call, including result construction and disposal. Variants share the eager-final baseline, current master adapter and unchanged optimized numerical dependencies; only receipt-capture allocation changes. Symbolica masters remain selected; N5 retry is inactive.

Table 10: **All fourteen workloads are retained.** Ratios below one favor lazy capture. Each cell shows the paired latency ratio, its 99% bootstrap interval and the unchanged per-family verdict.

Workload	Batch 1	Batch 32
Bubble · mixed Dual	0.9945 99% CI 0.9932–0.9969 Gain	0.9996 99% CI 0.9958–1.0021 Compatible with parity
Bubble · ordinary	0.9913 99% CI 0.9871–0.9976 Gain	0.9957 99% CI 0.9935–0.9968 Gain
Bubble · complex / exact zeros	0.9972 99% CI 0.9953–0.9979 Gain	1.0008 99% CI 0.9987–1.0031 Compatible with parity
Bubble · real / exact zeros	0.9984 99% CI 0.9940–1.0045 Compatible with parity	1.0089 99% CI 1.0038–1.0159 Regression
ggH · mixed complex	0.9979 99% CI 0.9957–1.0019 Compatible with parity	0.9999 99% CI 0.9975–1.0017 Compatible with parity
ggH · real / exact zeros	0.9951 99% CI 0.9890–0.9973 Gain	1.0022 99% CI 1.0002–1.0061 Regression
Triangle · ordinary	1.0077 99% CI 1.0053–1.0108 Regression	1.0018 99% CI 0.9981–1.0034 Compatible with parity

Five gains, six parity results and three qualified regressions; no noisy intervals. No adoption. Regressions occur for the real exact-zero bubble at batch 32, real exact-zero ggH at batch 32, and ordinary triangle singleton. Every 99% interval meets the fixed 2% width gate. The all-family no-regression criterion fails; all families remain retained.

Fourteen optimized preflight pairs preserve 10,560 unique complex output slots and exact status/metadata/context/counters. The earlier 26 debug tests remain separate. CPU 254; five warm calls; 41 paired blocks with 21/20 alternating order; retained calibration ≥200 ms on the faster side; 20,000 paired bootstrap resamples; all 1,148 observations retained. Lane/output checks precede and follow timing; iterations unwrap the top-level result. The host is shared.

Scope: N2/N3 f64 and flattened mixed Dual only; no new TIRust/IREGI or TIRust/COLLIER gate. Earlier eager/core and slim-RealBall cohorts remain separate; their ratios cannot be multiplied. Source: `validation/lazy-receipt-performance/`.

19 Additional bounded correctness checks

These completed checks extend the evidence ledger without changing production or any measured timing row. Each private candidate retains its own source and dependency snapshot; the raw-master replay gains do not establish whole-reduction performance for a different candidate.

Bounded study	Observed result and remaining limit
Selected native A0–C0	Eight ordinary f64 N2/N3 rank0/1 families at batches 1/32 pass all 1584 complete tensor coefficients per backend against original IREGI and the Symbolica baseline. Maximum normalized errors are 3.38×10^{-13} against IREGI and 2.07×10^{-13} between backends. The 297 actual master rows retain all 1782 six-output coefficients. Private opt0 build; whole-call performance is unmeasured.
Lazy proof capture	26 tests pass: 12 receipt, 8 lifecycle and 6 accounting checks. Fourteen paired workloads match exact values, status, metadata and counters through seven calls per implementation. Capture is created only when evidence is needed and discarded after each call. This private debug qualification is separate from the later optimized timing table; no adoption.
COLLIER N4 extension	Eight fixed massive cases cover ranks0–3 at batches1/32. Both scalar rows are accepted, maximum normalized error 2.76×10^{-14}. All six higher-rank COLI rows retain `AccFlag=-1`, despite raw agreement within 2.39×10^{-12}. DD mode returns `ErrFlag=-10` for all eight; those arrays are invalid reference evidence. No inputs or thresholds changed; no timings were run.
Fused core CSE	The N2 rank1 native/OneLoop exports remain byte-identical: 1013/850 instructions and 237/147 slots. All 422 exact roots and 730 control checks pass under the structured-SSA assumption. With no structural saving, the candidate stops before numerical or timing work; other topologies remain untested.

19.1 Scope of the independent checks

The selected-native study also passes 34 adapter cases, ten unchanged master paths covering 33 cases and three unchanged whole-core paths. Repeated whole responses are retained; direct adapter repeats are executable assertions with the common output saved. Initial build and fixture failures remain in the package. Native D0, high-point tensors and a production backend switch are outside this qualification.

The lazy-capture allocation occurs inside the existing argument-stage timer. The earlier eager allocation was outside that stage; the later paired comparison therefore uses total latency. The earlier final receipt cohort and its two qualified regressions remain unchanged.

N4 checks use 238 Symbolica-prepared polynomial arrays for metric traces, pinch translation, denominator-difference signs, UV finiteness and odd-time reflection. Their largest identity discrepancy is 2.89×10^{-12}; these constraints do not independently determine every spatial tensor component or override a rejected library status. The initial eight measured IREGI/COLLIER rows remain unchanged.

A retained **initial unmasked** thin-N6 rank2 probe passes seven exact/sidecar checks, then both certified-planar retries fail among six packed N5 children; that attempt produces no parent arrays. The later private mask followup has separate source, numerical and lifecycle qualification below. Neither changes the production N6/N7 domain or establishes performance parity.

PyO3 bindings and three reactive marimo notebook sources now follow `PLAN.md`'s pinned Community extension guide. Compile/type checks and notebook lint are distinct from installed-wheel execution; their bounded checkpoint appears separately. Native notebook runtime remains unqualified.

Source bindings, raw outputs, failures and reproduction commands are retained in `docs/data/results.json`. No elapsed correctness-run duration on this page is a benchmark. Production remains `06706e2` the optimized TIRust performance gate remains the measured `6b24c03` snapshot.

20 Additional execution qualification

20.1 Private N6 rank-two mask followup

Fourteen exact checks qualify the thin child assembly, full-shape zero-mask certificates, cache/context binding and route envelopes. Float128 and Float256, both requested 53, return all 60 native complex Laurent slots for the retained base point. Maximum normalized disagreement with original IREGI is 1.165×10^{-12} against the unchanged 10^{-10} gate; maximum 128/256 difference is 1.474×10^{-34} against 2^{-80}.

Check	Retained count	Scope
Precision comparison	2 × 60 slots	One retained base point at Float128/256; requested 53.
Lifecycle	11 calls / 17 accepted lanes	1020 complex slots; seven explicit rejection controls and two empty resets.
Singleton agreement	Zero differences	Values, precision and signed zeros agree with separate singleton calls.
Independent audit	327 bindings / 1140 slots	Saved-data arithmetic and source-archive audit; no kernel rerun.

The lifecycle corpus includes base/scaled points, mixed valid/invalid batches, reordering, chunks and resets. It shows isolation and absence of partial successful arrays; a rejected status alone does not prove the mathematical failure cause. The ordinary primal sensitivity estimate remains nonrigorous for compounded Taylor errors, and imported quadrature errors are unsupported.

Strict scope: private FromShifts/Preserve, unit powers, no dimension shift, N6 rank 2, native raw components. Numerical Dual, complex masses, N7, rank 7, generalized inputs, production adoption and performance remain unqualified. The earlier unmasked rejection and the 30 s exact-test timeout with zero completed tests remain preserved. The later unchanged 14-test gate passes within its separately declared 180 s bound. Sources: `validation/highpoint-thin-n6-mask/` and its independent checkpoint audit.

20.2 Private generic rank-three extension

A separate generic assembly passes 25 exact structural/integrand/context tests. At the retained N6 rank 3 base point, Float128/256 with requested 53 each return all 140 native complex Laurent slots, including 105 exact zero poles. Maximum original-IREGI discrepancy is 2.426×10^{-12} against 10^{-10}; precision difference is 1.474×10^{-34} against 2^{-80}. The exact tests cover structural identities through rank 7; numerical rank 7 is not qualified. This separate package adds no rank 3 lifecycle, numerical Dual, generalized/SuppliedG, N7, public adoption or performance result. The initial test compile failure and historical rank 7 timeouts remain. Source: `validation/highpoint-thin-n6-generic/`.

20.3 Initialization-order diagnostic

Two fresh processes retain 55 complete responses: seven control requests and 48 case requests. All numerical admissions pass; 37 within-process exact comparisons have zero differences. CASE-first reproduces all 32 saved case responses and all three controls exactly. CONTROL-first reproduces a one-ULP C0 difference and changes 24 finite imaginary tensor values per backend in the batch 32 rank-one triangle, by at most nine ULP (2.50×10^{-16}). The whole-case master captures and input/status/diagnostic records remain exact.

Registration/operand-order source facts suggest a mechanism, but no paired instruction exports establish the cause. This is an untimed, schedule-associated diagnostic, not universal determinism or an accuracy-threshold change. The failed v2 preflight remains unmeasured. A later four-engine cohort uses its separately reviewed CASE-first protocol; no diagnostic duration is a benchmark. Source: `validation/selected-native-initialization-order/`.

20.4 Python interface: historical compile and source typing

The PyO3 standalone/native crate passes Cargo checking with one shared Symbolica/Numerica/Graphica/PyO3/FeynKit owner identity. Its ordinary source consumer passes basedpyright with zero errors and warnings. This checks compilation and source-facing type signatures; installed-wheel typing and execution require the separate runtime evidence below. The initial build timeout and retained mypy/upstream-stub failures are preserved rather than counted as successful checks.

The initial notebook source capsule retains its original strict lint and nine display-helper tests. Later explorer, browser and native notebook checks are separately source-bound; historical results do not qualify a later source revision. Sources: `validation/python-interface/checkpoint-compile/` and `validation/python-notebooks/qualification.json`.

21 Python and notebook runtime

21.1 Inspected native wheel baseline

The exact baseline wheel builds, installs and passes shared-owner identity, packaged font/stub checks and installed-consumer typing. Of 29 focused tests, 28 pass; the real upstream graph renderer fails with `painted coordinates must be numbers`. Numerical, precision, Dual, dotted, batch, contraction and shared tensor/graph interoperability checks pass independently of that rendering failure. Tests use the actual shared upstream types from the installed native host.

Release-wheel performance, general mathematical-domain acceptance and full Community-host integration remain unqualified. Historical `6b24c03` measurements describe an earlier kernel. Source: `validation/python-interface/checkpoint-wheel-003/qualification.json`.

21.2 Actual native notebook execution: partial result

The inspected wheel is installed into the marimo 0.24.0 notebook environment. Four native test methods complete: two pass, two have three rendering-error occurrences. The successful controls check the dotted bubble against its mass Dual derivative, precision distinctions beyond binary64, failed-lane isolation and an edited-topology guard. Finite-difference values are displayed; this test adds no independent finite-difference acceptance gate.

Notebook	Actual checks	HTML export	Qualified scope
01: Graph → integral	Real renderer errors	21 cells / 1 error	Bubble, triangle Float and bubble DoubleFloat stop before their numerical assertions.
02: Dots and derivatives	Numerical controls pass	19 successful cells	Massive bubble identity, exact-input digits and invalid-neighbor masking.
03: Benchmarks	3 tests / 6 cohort choices	11 successful cells	Frozen evidence only; no numerical extension imported.

Both native exports and the original error remain archived. Rendering-dependent numerical assertions remain unrun; full gallery qualification remains open. Graph/math browser checks of the native notebooks are pending. Source: `validation/python-notebooks/native-wheel003/`.

21.3 Benchmark explorer and readable narrow charts

The separate explorer source revision passes strict lint and 15 evidence/display-helper tests, including all forty saved ratios and eighty CI endpoints. Actual marimo execution retains every cohort and its original short/noisy gates. Static Chromium desktop/narrow-dark checks preserve eight ratio cells and render without request, HTTP, console or page failures.

The chart now keeps a 930 px SVG inside controlled horizontal overflow, preserving readable labels within a 348 px narrow viewport; the document stays 390 px wide. Desktop cells and narrow/dark grid/chart crops were visually inspected. Earlier loader failure and small-chart screenshots remain historical. The export requires the official marimo CDN frontend and has no reactive backend; browser snapshots do not qualify numerical controls. Sources: `validation/python-notebooks/chart-readable/` and its separate source/runtime/browser predecessors.

22 Precision and high-point research

22.1 Typed direct E6: bounded common-mass fixtures

The unchanged private N2 direct genuine D6-pentagon engine is checked on common complex masses and zero invariants, with four- and six-component mixed Taylor shapes. Across ten groups, 65 lanes succeed and 65 intentionally reject; all 320 successful components satisfy the requested enclosure budget. Rejected lanes hide values. This extends the earlier one-axis prototype; it does not integrate the engine into the public reduction path.

Base type	Working / requested bits	Accepted / rejected lanes	Accepted components
f64	53 / 40	13 / 13	64
DoubleFloat	106 / 80	13 / 13	64
Float	128 / 80	13 / 13	64
Float	256 / 128	13 / 13	64
Float	384 / 128	13 / 13	64

Independent exact audit: 24,889 checks and 166 hashes, zero failures; the root independently reran that audit. Inputs distinguish 2^{-80} where the type permits it, retain mixed input precisions and verify scale-jet independence. Static Mixed4/Mixed6 and dynamic HyperDual metadata transport use Symbolica's Dualizer with NativeFloat workspaces; this is **not** `Workspace<StaticDual<T>>` qualification. Empty, odd, reordered, chunked, fresh, invalid-neighbor and independent concurrent workspaces pass compiled assertions; complete repeated arrays are not serialized.

Float256⟶384 arithmetic widths shrink for 102 positive coordinates and remain zero for 26. This comparison removes twice the retained truncation radius; it does not establish convergence of the fixed N2 mixed-cubic truncation. Instruction-lane visits are structural estimates. Unequal masses, nonzero invariants in this typed extension, generalized powers/dimensions, N4, contour/geometry certification and performance remain outside its scope. Production E6 failures remain unchanged. Source: `validation/e6-direct-n2-typed/`.

22.2 Exact child localization and mixed-jet counterexample

Two retained regular planar charts locate an ordinary child with Gram rank 2/determinant 0 and Cayley rank 4/determinant −4. Its multiplier vanishes at this exact ordinary input; generic symbolic aliasing still requests the singular PV child. This is preparation/identity evidence only: it does not identify the first nonfinite runtime instruction or evaluate a scalar master.

A separate exact mixed-jet counterexample has zero primal and first components but nonzero mixed cofactors −1/−1 and weights −2/−1 in the two charts. **Primal zero is insufficient for pruning.** Any future mask must certify the complete requested cofactor jet while retaining planar and chart regularity checks. No pruning implementation, successful N6 array, production change or performance result is claimed by these sources: `validation/highpoint-child-localization/`.

23 Independent pentagon adjudication

These checks establish selected mathematical values independently of an original-reference stability flag. They do not qualify every pentagon tensor, physical graph, runtime domain or numerical precision.

23.1 Rank-six Euclidean tensor

For squared masses $(4, 9, 16, 25, 36)$, $\mu = 3$ and numerator $q_0^5 q_3$, direct simplex integration gives native finite coefficient $-0.0024238378935447123i$. Independent coordinate mappings agree within 4.34×10^{-19}; reported quadrature error is at most 2.77×10^{-15}. The bound $F \geq 3.75$, angular integration and power counting exclude poles. This uses no reduction/master library; numerical error estimates are not interval certificates. See `validation/pv-high-rank/pentagon-simplex-oracle.json`. Historical binary64 errors remain 2.23×10^{-10} for TIRust and 3.40×10^{-10} for IREGI. Public rescue flags the primary unstable and verifies all 840 slots at 128/256 bits plus this coefficient below 10^{-13}. `validation/rescue-pentagon-rank6.json` retains the binary and assertions; full arrays were not printed and its exact build-source snapshot was not saved.

The vendored Symbolica baseline passes four scope tests, 57 core tests and all 1320 rank-seven slots at 128/256 bits, including the independent coefficient. Its source predates later precision/routing edits; its workspace compilation preserves 329 source/manifest hashes. The 109.44 s cold run, earlier 240 s timeout and 115.35 s isolated run are historical diagnostics. See `validation/symbolica-production-qualification.json`.

23.2 Physical canonical moment 54

For $gg \to HHH$, rank-four moment $q_0 q_3^3$ has UV degree -2; its UV pole must vanish. Original IBP instead returns approximately $3.33691766067 \times 10^{-6} i$ in native normalization. The 128/256-bit PV evaluation suppresses its OneLoop-normalized UV residual from 7.9×10^{-31} to 2.3×10^{-69}, with both IR slots zero. Its converged OneLoop finite coefficient is approximately $3.3474440772594743330 \times 10^{-13} + 2.9254797223941252086 \times 10^{-13} i$. This supports original PV/hybrid and rejects the original IBP result for this component. It is not a full physical tensor comparison. Raw precision evidence is in `validation/pv-high-rank/canonical-pentagon-precision.json canonical-pentagon-adjudication.json` records fixed-gate comparisons.

23.3 Standalone six-dimensional scalar

Use $J_1 = \frac{I_6}{i\pi^3}$, shifts $(0, e_0, e_1, e_2, e_3)$ and squared masses $(10, 12, 15, 18, 22)$. Here $F \geq \frac{39}{4}$, $\det R = 2600$ and $\det G = -16$.

Method	Finite J_1
Independent simplex	$-0.00017714013826400 81806076521310$
Original scalar/general APIs	-0.00014953025024128455; every mode reports stable

Increasing quadrature order and precision through 70 digits converges to the first value. Independent integration of ordinary D4 values over a common mass shift, with a separately integrated tail, agrees within 4.93×10^{-15}. A box-only half-line implementation also agrees to approximately 8.42×10^{-17} or better in the recorded checks. Ordinary D4 and coalesced D4/D6 normalization checks pass. Evidence and reproduction source are in `validation/shifted-pentagon/evidence.json`. Thirteen consolidated D6 tests cover eight genuine-master checks and five endpoints. Retained public f64 fixtures pass finite values, poles, scaling, rejected endpoints and lane isolation; the earlier quadrature-budget failure remains historical. Pre-vendor public DoubleFloat and Float128 tests now pass three scaled lanes at requested 80-bit accuracy and 10^{-23} normalized oracle tolerance. First common-mass Dual primal/derivative checks pass at 2×10^{-14}, with zero poles. DD/Float validation takes 165.69 s, exceeding its historical 150 s bound; this is not a benchmark. Build scope is in `validation/shifted-pentagon/public-core-qualification.json`. Paired vendored DD E6 passes exact allocation parity. Requested128 fails for the small-routing fixture; higher/mixed Dual, broader endpoints and native E6 remain unqualified. The original complex-box discrepancy is retained in `validation/shifted-pentagon/contours.json`.

A private input-enclosure extension passes five D0 pinches and two contractions at the single exact contour point $x = \frac{2}{5}$, Float256/request128: 40 rectangles and 20 rounded mass slots checked. Exact unshifted geometry remains assumed; no quadrature certificate or production E6 qualification follows.

24 Reference conventions

These conventions were read from the supplied original sources. Their presence in this report is not a claim that every adapter has passed its numerical tests.

24.1 Kinematics and scale

Use Minkowski signature $(+,-,-,-)$ with component order (p^0,p^1,p^2,p^3). Propagators are $D_j = \left(q+p_j\right)^2 - m_j^2 + i0$, raised to positive integer powers. Generalized integrals use $d = 4 + \mathrm{idim} - 2\varepsilon$ and the native raw measure; dimension shifts introduce powers of π. Native tensor reduction accepts one shift per propagator and removes the first common shift internally, then translates tensor coefficients back. The MadLoop interface instead receives the nonzero shifts and inserts a zero first shift. A reference comparison must preserve the chosen numerator routing when it changes the propagator direction.

The original `MU` / `MUR` argument is an energy scale, not its square. Its scalar wrapper passes μ^2 to QCDLoop and μ to OneLOop. Complex mass-squared inputs must retain their imaginary part; the reference switches to its OneLOop backend for complex masses. OneLoopMaster accepts real external squared invariants, complex squared masses with $\Im(m^2) \le 0$, and a positive real squared scale. Its continued logarithms and polylogarithms own the Feynman branch prescription; no independent branch implementation belongs in the reduction adapter. Preserve inputs by default; legacy on-shell snapping must be requested explicitly. Dual derivatives follow the selected analytic branch away from nondifferentiable boundaries.

24.2 Native poles and MadLoop conversion

The native result order is

$$(U,I,J,F) = (\varepsilon_{\mathrm{UV}}^{-1}, \varepsilon_{\mathrm{IR}}^{-1}, \varepsilon_{\mathrm{IR}}^{-2}, \mathrm{finite}).$$

These symbols label separate coefficients. In particular, a scaleless integral can have canceling UV and IR contributions even when its combined Laurent series is zero. Three combined coefficients do not determine their separate origin. OneLoopMaster uses $(\mathrm{finite}, \varepsilon^{-1}, \varepsilon^{-2})$ ordering.

The original `IMLOOP` adapter combines the native poles and maps a tensor of homogeneous rank r to MadLoop order (finite, simple pole, double pole). Defining $a = \gamma_E + \ln\pi$, $b = \frac{a^2}{2} + \frac{\pi^2}{12}$ and $c_r = \frac{(-1)^r}{i\pi^2}$, its explicit mapping is

$$(F_{\mathrm{ML}}, S_{\mathrm{ML}}, D_{\mathrm{ML}}) = c_r(F + a(U+I) + bJ, U + I + aJ, J).$$

For a OneLOop-normalized scalar master (L_0, L_1, L_2), the scalar wrappers use

$$J = i\pi^2 L_2, \quad U + I = i\pi^2(L_1 - aL_2),$$

$$F = i\pi^2\left(L_0 - aL_1 + \left(\frac{a^2}{2} - \frac{\pi^2}{12}\right)L_2\right).$$

The UV/IR split requires independent integral classification. In particular, the scaleless bubble is $(U,I,J,F) = (i\pi^2, -i\pi^2, 0, 0)$ in the original wrapper. Applying the MadLoop conversion to an already normalized master result would count normalization twice.

24.3 Polynomial and tensor interfaces

A rank bound R has $\binom{R+4}{4}$ Cartesian coefficients in the original four-component basis. `ti_reduce::calc_pos` traverses total degree in ascending order, then exponents of q^0, q^1, q^2 in descending order; the exponent of q^3 is the remaining degree. Thus the start is $\left(1, q^0, q^1, q^2, q^3, (q^0)^2, q^0q^1, q^0q^2, q^0q^3, (q^1)^2, \ldots\right)$. Numerator preparation and tensor output must share this order. The original native and MadLoop tensor conventions differ in multinomial factors: the native path multiplies symmetric components by the factor returned by `all_integers`, whereas `ML5_CONVENTION=.TRUE.` uses unweighted components. The independent oracle sets this flag **without** applying `IMLOOP`, preserving the native four-pole normalization. Mixed-monomial tests detect multiplicity mismatches.

Reduction modes are 0: IBP, 1: PaVe, 2: PaVe with IBP stability improvement. Direction mode 2 reverses the propagator order and shifts the loop origin. The source labels direction modes 4/5 as unimplemented quadruple precision; their numbers are not evidence of high-precision support.

25 Algebra reuse audit

All symbolic algebra belongs to Symbolica and the FeynKit / Spenso / Idenso ecosystem. Before introducing a missing operation, record three checks: Symbolica APIs and implementation; Symbolica usage/examples/tests; and FeynKit/gammaloop facilities and tests. An adapter or data-layout conversion must not grow an independent polynomial, simplification or matrix engine.

25.1 Existing facilities to reuse

- **Tensor identities and contractions:** Idenso contracts metrics and reduces supported gamma/color identities; Spenso carries tensor structure. Keep Lorentz dimension and any gamma-five prescription explicit.
- **Coefficient extraction:** FeynKit's tensor reducer uses Idenso contraction without global expansion, `to_dots()` and the shared `coefficient_list` collector. Preserve scalar factorization until polynomial extraction requires expansion.
- **Propagator and scalar-product algebra:** FeynKit's `IntegralFamily` uses Symbolica for polynomial conversion, exact rank and linear solves, basis completion and mappings. Its Gaussian-rational conversion avoids statistical expression-field zero tests that may discard small coefficients.
- **Numerical preparation:** Compile reusable Symbolica expressions outside the sample loop. Evaluation and scalar-master types must preserve the selected precision and all Dual components without binary64 round trips.
- **Laurent coefficients:** IBP/PV factors and native normalization use cached Symbolica `series` templates and `coefficient` extraction. The three reuse checks are the pinned Symbolica `src/atom/core.rs` and `src/poly/series.rs` APIs; its `src/derivative.rs` pole-series tests; and existing use in `vakint/src/projection.rs` and OneLoopMaster's `src/sheet_exact.rs`. No separate Laurent convolution engine is needed. Symbolica also evaluates the independent Gamma/series triangle oracle.

25.2 HEPKit tensor-reduction optimizations

The local `feynkit-tensor` reducer stores repeated-vector contractions as symmetry orbits instead of factorially many labeled pairings. Its joint-color dynamic program builds orbit intersections; alternating cycles build orbit Gram matrices; Symbolica solves the smaller orbit system. A rank-20 isotropic projection can have one orbit instead of 19!! labeled pairings.

The implementation caches canonical Weingarten coefficients and has a closed isotropic path. General projectors are validated through rank 20; the special all-equal isotropic route reaches rank 32. These are upstream limits, not a statement of iregi-rust's current supported domain.

For denominators with external momenta, declare an independent external basis and average only the transverse space. Odd total tensor ranks need not vanish. Keep D symbolic while constructing high-rank projectors; early integer specialization can make the universal metric basis singular. External Gram inverses can amplify small-Gram numerical errors, so projector preprocessing requires a conditioning-aware comparison against direct recurrences.

This projector reduces tensor numerators, not complete scalar integral families. Reuse it where applicable and benchmark the resulting plan; no factorial-count argument alone establishes a numerical speedup. Retain explicit output/pairing limits and report their failures honestly.

25.3 References and provenance

The untouched generalized IREGI entry point stops during factorial-table initialization: the configured powers/rank request factorial 16, while the helper rejects arguments above 12. A narrow repair in a generated reference copy replaces that table expression by its exact binomial recurrence. The untouched/repaired twins agree exactly on 24 ordinary cases; all 80 table entries match independent integer binomials; seven generalized fixtures check normalization, unit powers, mass derivatives and a tadpole dimension shift. The original `STOP` returns exit code zero, so the harness also requires an explicit result record. Evidence lives in `validation/reference/fixtures.json` and the report's JSON snapshot.

OneLoopMaster documents corrected analytic-continuation defects in its upstream OneLOop reference, including complex-mass box cases. Preserve each disagreement as a visible case and adjudicate it with independent analytic, high-precision or parameter-integral evidence. Agreement must not mean reproducing a known incorrect branch.

Primary references: Alwall et al., appendix C.1 HEPKit TensorReducer Goode et al., tensor projectors Collins–Matsumoto, Weingarten recurrence NinjaRS report style. Local source audit: FeynKit's `feynkit-tensor/src` projector modules and `feynkit-graph/src/integrals.rs` original IREGI's `IREGI4ML5_interface.f90`, `ti_reduce.f90`, `si_reduce.f90`, `mis_warp.f90` and `funlib.f90`.